

ARTICLE

Open Access



Anticancer effects of aloe-emodin from *Rheum undulatum* L. through activation of the p53 pathway in human prostate cancer cells

Nguyen Khoi Song Tran^{1†}, Nhu Quynh Nguyen^{2†}, Sullim Lee³, Seung Hyun Kim⁴, Daesik Jeong⁵, Eunjeong Seo^{6,7}, Jin Ju Park^{6,7}, Jaejin Cho^{6,7*} and Ki Sung Kang^{2*} 

Abstract

Aloe-emodin, an anthraquinone compound naturally derived from *Rheum undulatum* L., has gained extensive research attention owing to its various pharmacological effects, including its potential as an anticancer, antiviral, anti-inflammatory, antibacterial, and anti-parasitic agent. It has demonstrated notable inhibitory effects against various types of cancer and cancer cells. Prostate cancer is among the most commonly identified cancers globally and remains a leading cause of cancer-associated deaths in men, often presenting challenges in early detection due to its asymptomatic nature during initial stages. The aim of present study was to determine the biological activity of aloe-emodin obtained from *Rheum undulatum* L. involving activation of the p53-dependent pathway in certain human prostate cancer cell lines. We explored the mechanisms underlying the anticancer effects of aloe-emodin using LNCaP cells, which include p53-wild type and phosphatase and tensin homolog-deficient mutated genes, a widely studied model in genomic research. Aloe-emodin induced apoptosis in LNCaP cells through several mechanisms, including upregulation of the cleavage of caspase-8 (a cross-linked promoter of cell death signals), phosphorylation of p53 at serine 15, DNA fragmentation, cleavage of poly [ADP-ribose] polymerase, and promotion of cell death. These findings strongly indicated that aloe-emodin's anticancer properties in human prostate cancer involve the activation of p53-induced cellular senescence. Conclusively, the findings of this study imply that aloe-emodin extracted from *Rheum undulatum* L. is a potential therapeutic compound for adjuvant chemotherapy that induces apoptosis and pyroptosis, an innate immune response, in preventing the progression of precancerous lesions in patients with prostate cancer.

Keywords Aloe-emodin, *Rheum undulatum* L., Anti-human prostate cancer, P53, Caspase-8, Programed cell death

[†]Nguyen Khoi Song Tran and Nhu Quynh Nguyen have contributed equally to this work.

*Correspondence:

Jaejin Cho

jcho@snu.ac.kr

Ki Sung Kang

kkang@gachon.ac.kr

Full list of author information is available at the end of the article

Introduction

Rheum undulatum L., a plant native to Siberia and China [1], has been extensively studied for its pharmacological properties, which encompass potential benefits as an anticancer, anti-allergy, anti-obesity, and anti-platelet aggregation agent [2, 3]. Among the constituents of medicinal herbs in the *Rheum* genus, including *R. undulatum*, anthraquinone and stilbene derivatives are major ones [4, 5]. Anthraquinones, especially, have exhibited promising therapeutic features in cancer treatment [6]. Notably, aloe-emodin was identified as a component with significant curing potential out of the anthraquinones derivatives of *R. undulatum* [7].

Aloe-emodin, also known as 1,8-dihydroxy-3-(hydroxymethyl) anthraquinone, is a natural origin anthraquinone compound found in several traditional remedial plants such as *Aloe vera* [8], *Rheum emodi* [9], and *Rheum undulatum* [7]. Its pharmacological properties have been extensively investigated, revealing a wide range of potential benefits inclusive of anti-inflammatory, anticancer, antiparasitic, neuroprotective, and hepatoprotective effects [10]. Currently, it has gained acceptance for the notable anticancer effects on various tumor cells and cancer types, such as bladder, cervical, gastric, colorectal, glioma, lung, melanoma, and hepatoma cancer [11]. Despite the advance that aloe-emodin has been demonstrated to trigger mitochondrial dysfunction and pyroptosis by activating the Caspase-9/3/Gasdermin E pathway in HeLa cells as well as inhibit prostate cancer growth by directly initiating the mTOR complex 2 [12, 13], the comprehension of crosstalk between various programmed cell death mechanisms is incomplete. Further research into the therapeutic potential and molecular mechanism of action of aloe-emodin is crucial to develop targeted therapies for cancer and other related conditions.

In light of aloe-emodin's established anti-cancer properties, we conducted a gene screening analysis using Enrichr software to uncover potential mechanisms associated with cancers. The analysis revealed specific pathways and mechanisms influenced by aloe-emodin that impact the proliferation of cancer cells (Table 1). These pathways encompass cell proliferation, regulation of the cell cycle, and pathways involved in cell death such as p53, TNF, VEGF, and apoptosis. The findings denote that these pathways may play pivotal roles in the suppression of cancer cells. Prior studies have highlighted that aloe-emodin inhibits the growth of diverse cancer cell types through cell cycle arrest, particularly at the G2/M checkpoint. This effect has been observed in gastric cancer SGC-7901 cells [14], leukemia HL-60 cells [15], colon cancer SW 480 cells [16], and cervical cancer HeLa cells [17].

Cancer, one of modern deadly diseases, is recognized not just as a singular, but as a complex and diverse collection of disorders, each with its unique biological and clinical characteristics. The cancer therapies research propose many approaching directions including traditional and modern medicines [18–20]. Prostate cancer represents a serious global health concern, being the most regular and leading cause of male cancer-associated deaths worldwide [21]. Early stages of the disease often present without symptoms, complicating timely diagnosis [22]. While many cases progress slowly, prostate cancer cells can metastasize via hematogenous routes, predominantly affecting sites including the adrenals (13%), pleura (21%), liver (25%), lung (46%), and bone (90%) [23]. Treatment strategies depend majorly on cancer cell sensitivity to male reproductive hormones and the extent of metastasis. Current therapies, including surgery and radiation, can impact patients' quality of life while their effectiveness is limited by metastatic activities [24]. Additionally, prostate cancer frequently exhibits resistance to multiple chemotherapy drugs, encompassing various classes of cytotoxicity agents [25]. Consequently, there is an urgent demand for interventions that can effectively suppress early-stage tumor growth and introduce novel chemotherapy approaches to combat this challenging disease.

Cellular survival and resistance to cell death are critical factors in the maturing and treatment of cancer. Cancer cells' ability to evade cell death not only initiates the disease but also contributes to treatment resistance, relapse, and metastasis. Modern cancer treatment strategies such as radiotherapy, chemotherapy, and immunotherapy aim to maximize cancer cell destruction while minimizing damage to healthy tissues [26]. Apoptosis, an essential process in cell death, occurs through two primary pathways: the extrinsic and intrinsic pathways. The pathways that are activated by ligands such as tumor necrosis factor α (TNF α), TNF-related apoptosis-inducing ligand (TRAIL), and Fas ligand (FasL) named extrinsic pathways. These binding complexes trigger the activation of caspase-8 initiating apoptosis. In contrast, the intrinsic pathway responds to internal cellular damage, typically involving mitochondrial dysfunction. This pathway is administrated by Bcl-2 family proteins, leading to cytochrome c (Cyt-C) and SMAC/DIABLO release. Cyt-C activates caspase-9 through Apaf-1 proteins, culminating in apoptosis of cancer cells [27].

Caspase-8 plays a critical role in triggering apoptosis through death receptor-mediated and mitochondria-dependent pathways via cleaving proapoptotic BH3 interacting domain death agonist (Bid). Caspase-9, another cell death initiator caspase, was observed for ability of inducing proteolysis certain important cellular

Table 1 Pharmacological activity and mechanisms of aloe-emodin

Cancer	Cell line/model	Activity/mechanism of action	References
Lung	CH27	Bax and Fas signaling pathways	[84]
	H460	Caspase-3-dependent pathways	[85, 86]
	SCC	MAPKs and PI3K/AKT	[87]
Tongue	SCC-4	Downregulation of ATM, ATR, BRCA1, and MGMT	[11, 88]
Esophagus	TE-1	Inhibits AKT and ERK	[89]
Gastric	AGS and NCI-N87	Stimulate the cleavage of caspase-3 and inhibit the casein kinase II activity	[90]
	MGC-803	Decreases ALP activity	[91]
	SGC-7901	PKC pathway and c-Myc suppression; mediates apoptotic cell death and necrosis; the intrinsic pathway	[11, 14, 90]
Liver	MKN45	Mediates the enrichment of G0/G1 checkpoint	[92]
	HepG2	Upregulation of Fas/APO1 receptor and Bax	[93–95]
	Hep3B	Induces p21 expression	[90, 96]
	Huh-7	Suppresses CAPN2 and UBE3A protein expressions	[97]
Breast	MCF-7	Suppresses cell proliferation, adhesion, invasion, and mediates apoptosis. Suppresses the expression of ER α	[98–100]
Melanoma	HER-2	Induces the expression of protein related to Akt/mTOR signaling pathway	[101]
	A375.S2	Suppresses the expression of mRNA and the activity of NAT enzyme	[102]
	B16	Activates ERK1/2	[103]
	A375	Inhibits ERK	[103]
	B16.F10	Suppresses MMP-9 activity	[104]
Leukemia	SK-MEL-110 and SK-MEL-28	Upregulates inflammation-associated factor expression	[105]
	HL60	Activates caspase-3 activity	[15]
	U937	Enhances MMP secretion	[106]
Colon	WiDr	Suppresses the translocation of nuclear factor of NF- κ B and its DNA binding	[107]
	SW840 and SW620	Suppresses ERK1/2 phosphorylation and induces the activity of	[16]
	Apc-deficient Min mouse model	MAPK/JNK and p38 MAPK Suppresses cell proliferation	[108]
Cervical	HeLa	Mediates the accumulation of G2/M and S checkpoints and increases the activity of ALP	[17, 109]
Glioblastoma	U-373MG	PKC signaling pathway	[110]
	C6 glioma	Suppresses ERK1/2 protein expression	[111]
	U87	The mitochondrial pathway	[112]
Ovarian	HO-8910PM	Downregulates the protein and mRNA expression of FAK	[113]
Bladder	T24 bladder	Induces p53 and caspase-3 expressions	[114]
Head and neck	NPC	The mitochondria pathway	[115]
Prostate	PC-3	mTORC2 signaling pathway	[13]
Bone (Osteosarcoma)	MG-63	Targets ROS-JNK	[116]

proteins involving in cell cycle and DNA metabolism by activating effector caspase-3 and caspase-7 [28]. In fact, activation of caspase-8 predominantly occurs through death receptor signaling, while caspase-9 initiation is more reliant on the mitochondrial pathway. These initiating caspases trigger downstream caspases such as caspases-3, 6, 7, leading to the cleavage of various cellular components involved in apoptosis [29]. Inhibition of TAK1 results in caspase-8 activation within murine macrophages that leads to the cleavage of gasdermin D (GSDMD) as well as gasdermin E (GSDME),

ultimately causing pyroptosis [30]. This programmed cell death potentially initiates the release of cytokines, activate macrophages and T lymphocytes, inducing a strong immune phagocytosis in place [31]. Notably, drug-induced caspase-8 activation can occur through both the death receptor pathway and the mitochondrial dependent pathway [32]. Lowering caspase-8 levels leads to G2/M cell cycle arrest, stabilizes p53, and triggers p53-dependent intrinsic apoptosis within tumor cells, specifically at the G2/M checkpoint [33]. Additionally, the activation of caspase-9 was proved to take

part in pyroptosis by mediating the activation of caspase-3 and 6 which in turn cleaved GSDMC [34, 35].

Targeting p53 holds great promise for prostate cancer gene therapy [36]. The p53 pathway is triggered by stress factors that impact DNA replication and cell division. Upon cellular stress, phosphorylation of p53 at serine 15 occurs, enabling a transcriptional activation, regulating cell cycle arrest, cellular senescence, or apoptosis [37]. Clinical trials were being conducted on various anticancer drugs which were designed to induce cellular senescence in both cancer cell lines and tumors by modifying the p53 status [38]. In addition, both p53 and p73 protein can function as upstream controllers of caspase-8. Caspase-8 in turn is an important intermediary in the p53/p73 apoptotic pathway triggered by etoposide in chemoresistant caspase-8 deficient HOC313 cell [29]. Another study also revealed that p53 stimulated the GSDME expression by interacting with a dedicated p53 binding site within the pyroptotic GSDME gene [39].

As reported previously, aloe-emodin was assessed as chemotherapy in the various kinds of cells and addressed apoptosis and cell cycle arrest pathways [40]. To further elucidate its mechanism, this study examined the impact of aloe-emodin on cell viability and p53 functionality across three metastatic prostate cancer cell lines: p53-deficient PC-3, p53-mutated DU-145, and p53 wild-type LNCaP cells. Subsequently, the study assessed the indicators of p53-dependent cell death in LNCaP cells, including apoptotic cell percentage, DNA fragmentation, and nuclear fragmentation using image-based flow cytometry and Hoechst staining. We aimed to investigate how aloe-emodin extracted from *R. undulatum* L. influences the expression of tumor protein p53 and the regulation of this compound on cellular senescence, apoptosis as well as pyroptosis—a novel cancer therapeutic direction—in human metastasis-derived prostate cancer cells.

Our findings suggested that caspase-8-mediated p53-dependent cell death is a key mechanism underlying the *in vitro* proliferation inhibitory effect of aloe-emodin on human prostate cancer cells. This highlights the potential of aloe-emodin as a promising adjuvant therapy for prostate cancer patients.

Materials and methods

Preparation of *R. undulatum* L. and compounds extraction

The chemical structure of aloe-emodin was confirmed by NMR and high-resolution electrospray ionization-mass spectra as reported previously [41].

Related signaling pathways

The related signaling pathways of target genes were predicted by Enrichr software [42], referring to the KEGG_2019_Mouse gene-set library [43]. The adjusted

p-values of the related signaling pathways in this study were calculated, $p < 0.05$ was considered statistically significant.

Cell viability assay

The cells were seeded on 96-well plates at a density of 5×10^3 cells/well for 24 h and exposed to 50 and 100 μM of aloe-emodin and cisplatin in a time-dependent experiment (24 h, 48 h, and 72 h). Cell proliferation efficiency and half maximal inhibitory concentration (IC_{50}) was evaluated as reported previously [44].

Image-based flow cytometry assay

LNCaP cells were incubated on 6-well plates at a density of 5×10^5 cells/well for 24 h. The cells were then supplied with fresh medium including 20, 50, and 100 μM of aloe-emodin and incubated for an additional 24 h. The apoptotic cell level was assessed using Invitrogen Tali image-based cytometer (Life Technologies, Tokyo, Japan). As reported previously [45].

Immunoblotting analysis

To initiate western blotting, LNCaP cells were first seeded on 60 mm plates at a density of 4×10^5 cells/well for 24 h. Immunoblotting analysis was conducted as reported previously [44].

Hoechst staining assay

LNCaP cells were plated on eight-well chamber slides (SPL Life Sciences, Orlando, FL, USA) with a population of 5×10^4 cells/well for 24 h. Hoechst stain was carried out as reported previously [45].

Statistical analysis

Data in triplicate are performed as the means $\pm$ standard error (SEM). Tukey's honestly significant difference for one-way analysis of variance (ANOVA) was employed for statistical analysis using the R program (version 3.3.3). Statistical significance is valued when $p < 0.05$.

Results

Predicted target genes and their related signaling pathways for bioactivity of aloe-emodin

Enrichr software was employed to identify the target genes affected by aloe-emodin. The software categorized these target genes into specific signaling pathways (Table 2). Remarkably, these pathways are closely linked to reproductive cancer and steroids. Furthermore, the analysis highlighted pathways related to critical cellular processes: cell proliferation, cell cycle regulation, and cell death, and those involving p53, TNF, VEGF, and apoptosis. These pathways are anticipated to play a crucial role in the suppression of cancer cells by aloe-emodin.

Table 2 Predicted target genes and signaling pathways for the bioactivity of aloe-emodin

Pathways	Overlap	Genes	Number of target genes	Number of target genes/group
Pathway in cancer	14/535	GSK3β, HSP90αA1, NOS2, PTGS2, ESR1, ESR2, CASP9, AR, CASP8, CASP3, CDK2, PIM1, PPARγ, PRKACα	14	14
Proteoglycans in cancer	5/203	CASP3, MAPK14, PRKACα, ESR1, TNF	5	5
Reproductive cancers				
Prostate cancer	5/97	CASP9, AR, GSK3β, HSP90αA1, CDK2	5	7
Breast cancer	3/147	GSK3β, ESR1, ESR2	3	
Endometrial cancer	2/58	CASP9, GSK3β	2	
Cell proliferation, cell cycle, and cell death				
p53 signaling pathway	6/71	CASP9, CASP8, CASP3, CDK2, CHEK1, CDK1,	6	13
TNF signaling pathway	6/110	CASP8, IL1β, CASP3, MAPK14, PTGS2, TNF	6	
VEGF signaling pathway	4/58	CASP9, NOS3, MAPK14, PTGS2	4	
Apoptosis	5/141	CASP9, CASP8, CASP3, EIF2S1, TNF	5	
Necroptosis	4/176	HSP90αA1, CASP8, IL1β, TNF	4	
Reproductive steroids				
Progesterone-mediated oocyte maturation	6/90	CCNA2, HSP90αA1, CDK2, CDK1, MAPK14, PRKACα	6	11
Estrogen signaling pathway	6/134	NCOA2, HSP90αA1, NOS3, PRKACα, ESR1, ESR2	6	
Ovarian steroidogenesis	2/57	PTGS2, PRKACα	2	
Other				
Neurotrophic signaling pathway	2/121	GSK3β, MAPK14	2	2

Among the predicted reproductive cancers, prostate cancer exhibited the highest representation, with a notable overlap of 5/97 genes. In the context of cell cycle, cell proliferation, and cell death influenced by aloe-emodin, the p53 signaling pathway emerged as the most prominent, with an overlap of 6/71 genes. This suggests that aloe-emodin’s potential in suppressing cancer cells may primarily involve these specific pathways and, notably, p53, which is a key manager of cell cycle and apoptosis. Additionally, we found that caspase-8 presents in most of cell proliferation, cell cycle, cell death, and cancer pathways which confirms the multidimensional potential of this caspase in cancer therapies.

Aloe-emodin compound isolated from *R. undulatum* L.
Dried rhizomes of *R. undulatum* L. were subjected to the extraction and solvent partitioning, resulting in three crude fractions: CHCl₃, EtOAc, and H₂O. Within the EtOAc fractions of *R. undulatum* L., four distinct compounds (C1–C4) were isolated and identified. Using high-resolution electrospray ionization mass spectrometry (HR-ESI-MS), along with in-depth 1D and 2D nuclear magnetic resonance (NMR) spectroscopic analysis as mentioned in the materials and methods, the isolated compounds were confirmed to be aloe-emodin (C1) (Fig. 1).

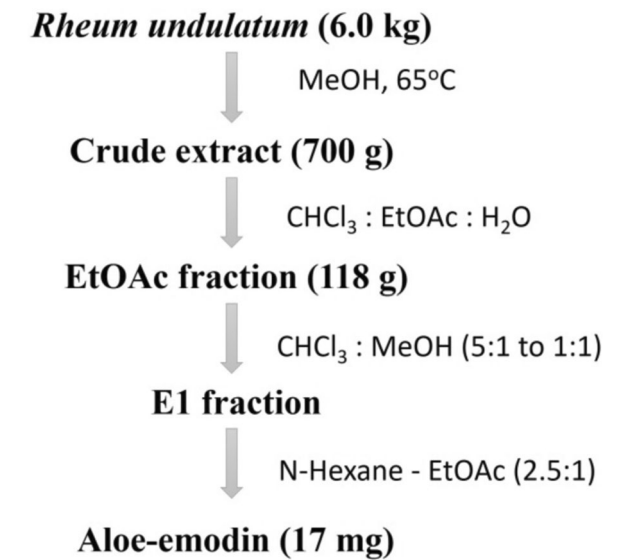


Fig. 1 The isolating and fractionating scheme of aloe-emodin from ethyl acetate fraction of *R. undulatum* L. The dried rhizomes of *R. undulatum* L. underwent extraction followed by solvent partitioning into CHCl₃, EtOAc, and H₂O fractions. The compounds isolated were confirmed as aloe-emodin through HR-ESI–MS and detailed analysis using NMR spectroscopy

Inhibition of aloe-emodin on the proliferation of human prostate cancer cell lines
Aloe-emodin showed inhibition on the proliferation of

PC-3, DU-145, and LNCaP cells which depended on both the dosage and duration of exposure. After 24 h incubation, the cell proliferation percentages at a 50 μ M dosage were $88.1 \pm 0.8\%$ for PC-3, $92.5 \pm 1.5\%$ for DU-145, and $85.2 \pm 1.8\%$ for LNCaP. At a concentration of 100 μ M, these percentages were $77.1 \pm 1.0\%$, $67.3 \pm 3.6\%$, and $70.3 \pm 4.2\%$, respectively, presenting significant inhibition of all three cell lines. However, the IC_{50} values for all cell lines exceeded 100 μ M, with no substantial differences among them (Fig. 2A).

Following 48 h of incubation, the proliferation of all three cell lines notably decreased at 50 and 100 μ M concentrations of aloe-emodin. PC-3 and DU-145 cells exhibited IC_{50} values exceeding 100 μ M, with viabilities of $73.0 \pm 2.3\%$ and $64.3 \pm 1.5\%$ at 100 μ M, respectively. However, the p53 positive LNCaP cells demonstrated a viability of $49.8 \pm 2.8\%$ at 100 μ M, with an IC_{50} of 99.67 ± 2.0 μ M. After 72 h of treatment, PC-3's IC_{50} remained above 100 μ M, while DU-145 and LNCaP cells had IC_{50} values of 88.6 ± 3.8 μ M and 73.4 ± 1.9 μ M, respectively.

Cisplatin served as a reference drug, with IC_{50} values for PC-3 exceeding 100 μ M at all-time points. For

DU-145 cells, the IC_{50} values were 89.7 ± 2.0 μ M at 48 h and 68.2 ± 1.8 μ M at 72 h. Similarly, LNCaP cells displayed IC_{50} values for cisplatin treatment of 69.6 ± 1.5 μ M at 48 h and 38.9 ± 0.6 μ M at 72 h (Fig. 2B).

Effect of aloe-emodin on p53-mediated expression of proteins related to programmed cell deaths in human prostate carcinoma PC-3, DU-145, and LNCaP cells

Exposure of PC-3, DU-145, and LNCaP cells to aloe-emodin led to a decrease in cell proliferation that correlated with the concentration used. To assess the involvement of programmed cell death in cell number suppression, we analysed the expression levels of cleaved Poly (ADP-ribose) polymerase (c-PARP), and p-p53 (serine 15). To investigate the role of cell death mediators and caspases inducers in cell suppression, p21, cleaved caspase-8 and cleaved caspase-9, Bax, and apoptotic regulator B-cell lymphoma 2 family (Bcl-2) protein expression levels were examined.

Notably, p53-mutant DU-145 cells exhibited a c-PARP expression profile which is similar to that of p53 wild-type LNCaP cells, suggesting caspase-mediated cell death in both. In contrast, PC-3 cells lacking p53 did not exhibit

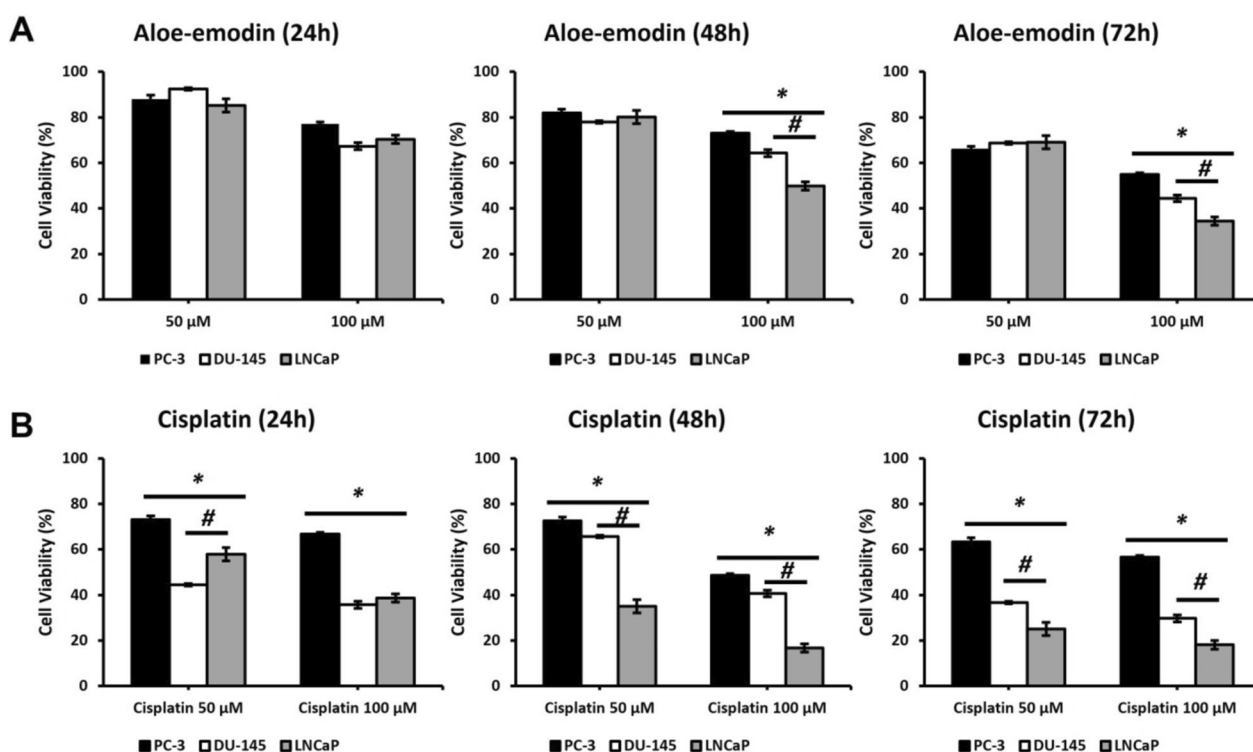


Fig. 2 Effect of aloe-emodin isolated from *R. undulatum* L. on the proliferation of human prostate cancer cells. PC-3, DU-145, and LNCaP cells were incubated in 96-well plates with a density of 5×10^3 cells/well for 24 h. Cells were exposed to 50 and 100 μ M of aloe-emodin (A) and cisplatin (B), respectively. Alive cell's number was examined using cell viability assay kit (Ez-Cytox) for 24 h, 48 h, and 72 h post-treatment. Data are performed as Mean $\pm$ Standard Error of Mean (SEM). (*) $p < 0.05$ is significant difference of LNCaP versus PC-3 cell and (#) $p < 0.05$ is significance of LNCaP versus DU-145 cell at the same concentration

measurable levels of c-PARP. However, after a 48 h exposure to aloe-emodin, no significant impact on the full-length PARP in all three cell lines was observed (Fig. 3).

In this study, aloe-emodin significantly altered the expression of specific cellular markers in LNCaP cells after a 48 h incubation period. Particularly, p-p53, a phosphorylated form of the p53 protein at serine 15, was upregulated, signifying an activation of the p53 pathway associated with cell cycle arrest. In addition, the activation of cleaved caspase-8 and 9, two key enzymes involved in both apoptosis and pyroptosis, suggests the initiation of programmed cell deaths (Fig. 4B). Although there is no significant difference was found in protein expression levels of p21 and Bax in aloe-emodin treated group, we observed a reduction in Bcl-2 levels, an

anti-apoptotic protein, indicating a shift in the cellular balance toward promoting apoptosis (Fig. 4A). The Bax/Bcl-2 ratio indicates cell death switch which is elevated in cell death conditions as well. These changes collectively point to the induction of apoptotic characteristics in LNCaP cells upon exposure to aloe-emodin.

Effect of aloe-emodin on p53-mediated apoptotic cell death in LNCaP cells

Following a 24 h exposure, aloe-emodin demonstrated the ability to elevate the proportion of cell death in LNCaP cells dose-dependently. The images obtained from an image-based cytometry assay (Fig. 5A) visually revealed an increase in apoptotic bodies within the treated groups using Annexin V staining, especially with

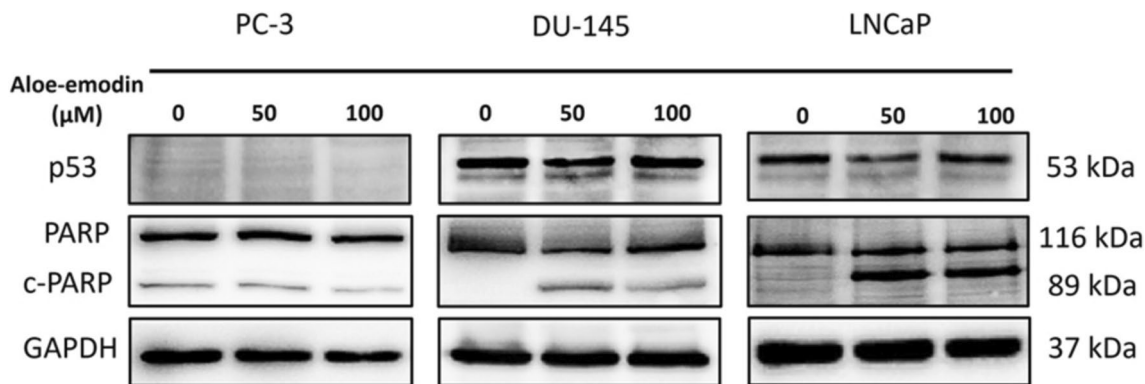


Fig. 3 Effect of aloe-emodin on protein expression of p53 in apoptosis in PC, DU-145, and LNCaP cells. PC, DU-145, and LNCaP cells were exposed to aloe-emodin at vary concentrations (50 and 100 μM) for 48 h, followed by extraction of total protein. To assess the role of apoptosis in cell suppression, western blotting was conducted to measure the p53 and c-PARP protein expression levels, which were referred to GAPDH

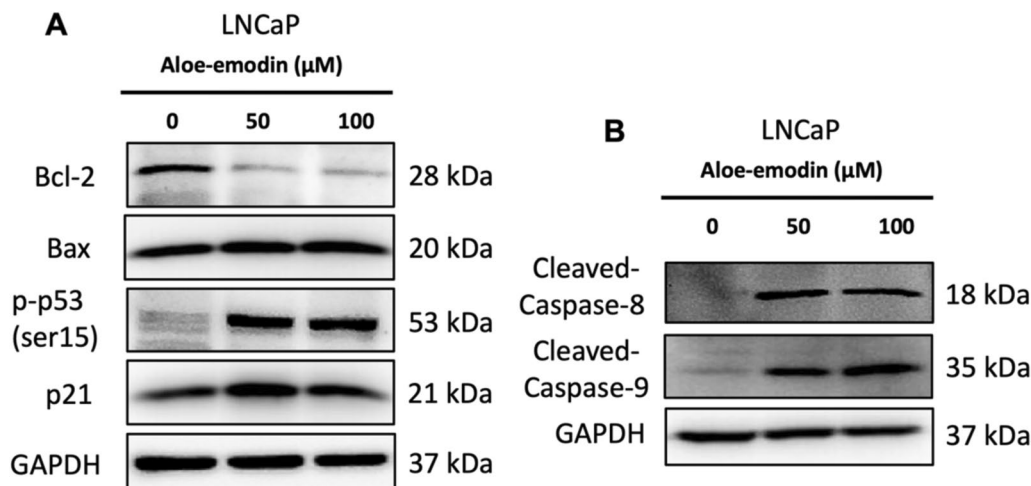


Fig. 4 The impact of aloe-emodin on expression of p53 phosphorylation and cell death mediators in LNCaP cell. LNCaP cells were exposed to various doses of aloe-emodin (50 and 100 μM) for 48 h and extracted for total protein. To examine the contribution of cell death mediators (A) and caspases (B) to cell suppression, western blotting for Bcl-2, Bax, p-p53 (serine 15), p21, cleaved-caspase-8, and cleaved-caspase-9 protein expression were conducted compared to GAPDH

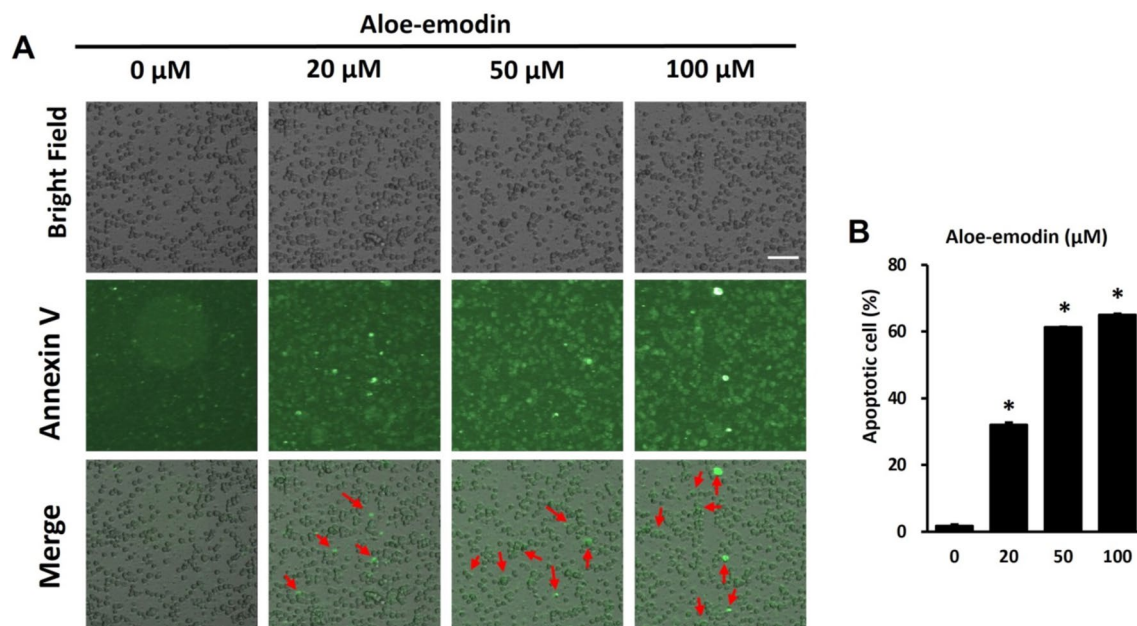


Fig. 5 Effect of aloe-emodin on the induction of programmed cell death in LNCaP. LNCaP cells at a density of 4×10^5 cells/well were plated in 6-well plates for 24 h and treated with aloe-emodin at concentrations of 0, 20, 50, and 100 μM . After an additional 24 h incubation period, the cells were collected, stained with Annexin V, and analysed using image-based cytometry assay. **A** Images captured during the image-based cytometry assay. Red arrows indicate apoptotic bodies. **B** The bar graph compares the percentage of apoptotic cells observed after 24 h of treatment with aloe-emodin. (*) $p < 0.0001$ is significant compared to the non-treated group. The data was presented as Mean $\pm$ Standard Error of Mean (SEM). Scale bar is 20 μm

50 and 100 μM concentration of aloe-emodin treatment, in comparison with the non-treated cells.

Aligning with the quantitative apoptotic cell percentage data (Fig. 5B), at concentrations of 20, 50, and 100 μM , the percentages of apoptotic cells were $32.0 \pm 0.7\%$, $61.3 \pm 0.1\%$, and $65.0 \pm 0.3\%$, respectively. In contrast, the non-treatment group had only $1.7 \pm 0.5\%$ dead cells, rendering these increases statistically significant ($p < 0.0001$).

Effect of aloe-emodin on the nuclear envelope fragmentation and DNA damage response in LNCaP cells

P53, a key regulator, influences various cellular processes, encompassing apoptosis, senescence, autophagy, and the DNA damage response (DDR) [46]. Notably, aloe-emodin treatment prompted distinctive changes in LNCaP cells after a 24 h exposure, as observed through Hoechst staining, unveiling the condensed DNA structures (Fig. 6). Microscopic bright field images further depicted evident cell damage (black arrows) in the treated groups. Magnified images confirm nuclear membrane fragmentation.

These findings signify the multifaceted impact of aloe-emodin on LNCaP cells, potentially involving p53-mediated processes. The observed changes in nuclear morphology, such as condensed DNA and nuclear membrane fragmentation (red arrows), suggest a disruption in cellular integrity.

Discussion

Prostate cancer ranks among the most prevalent and fatal cancers globally. The primary treatments currently in use are surgery and chemotherapy. Nevertheless, individuals with prostate cancer and their partners often experience disruptions in their daily lives and encounter psychological challenges post-surgery [47]. Therefore, research investigating the anti-cancer properties of specific natural extracts has been initiated, with a focus on prostate cancer. Our findings reveal that aloe-emodin's pharmacological network demonstrates inhibitory effects across various pathways implicated in numerous cancer types, including mTORC, MAPK, and caspase-dependent pathways (Table 1). These pathways are closely involved in controlling cell proliferation and cell cycle.

Utilizing Enrichr software, we conducted predictions regarding the signaling pathways targeted by aloe-emodin for its anti-cancer activity (Table 2). These predictions indicate a strong correlation of the pathways to reproductive cancer and reproductive steroids. Particularly, pathways crucial for cell proliferation, the cell cycle, and cell death—including TNF, VEGF, p53, and apoptosis—are expected to play a pivotal role in the suppression of cancer cells induced by aloe-emodin.

The p53 pathway has previously been established as playing a crucial role in cell regulation and tumor

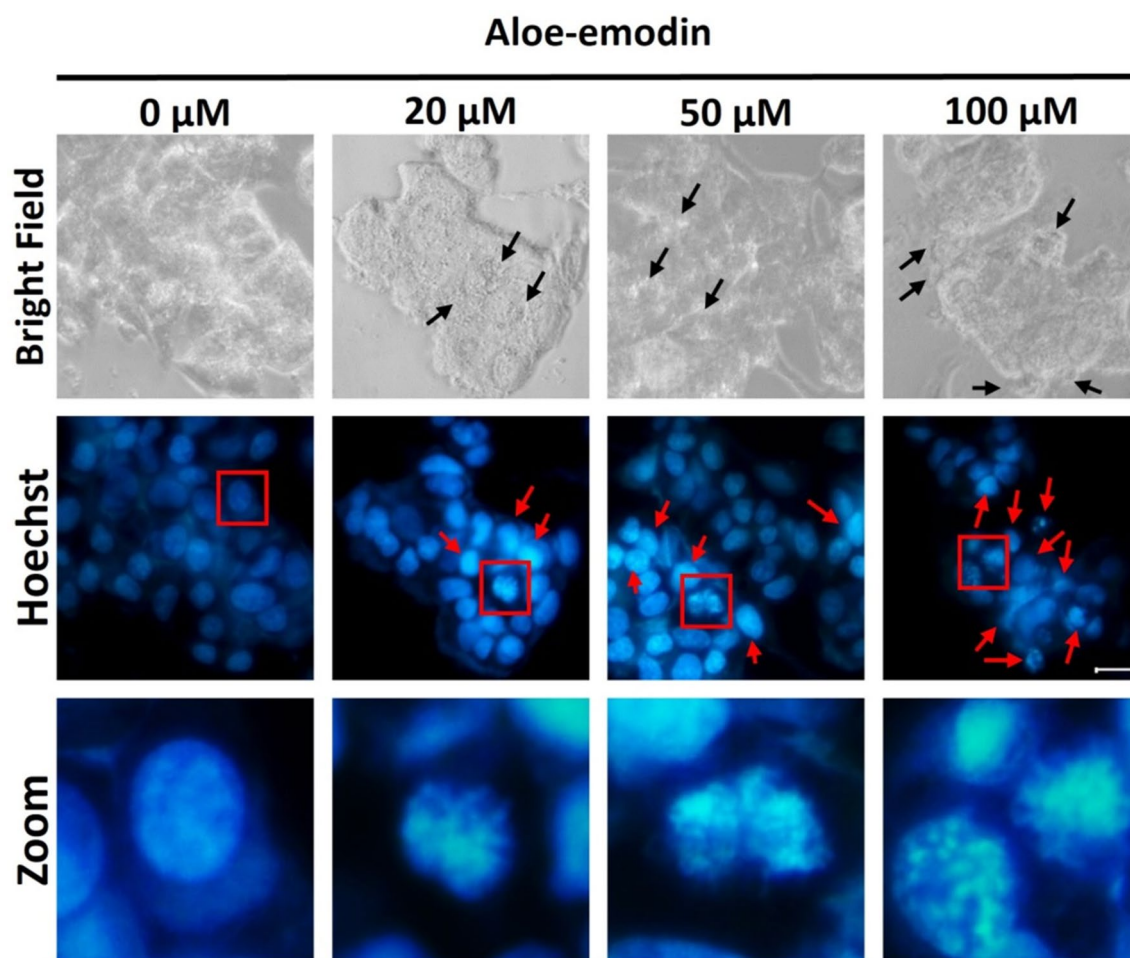


Fig. 6 Aloe-emodin mediated nuclear envelope fragmentation and DNA damage in LNCaP cells after 24 h of treatment LNCaP cells were incubated in an 8-well chamber at a density of 5×10^4 cells/well for 24 h. Afterwards, cells were treated with aloe-emodin at concentrations of 20, 50, and 100 μM for 48 h. **A** Cells were marked with Hoechst dye and examined under a fluorescent microscope. The scale bar represents 20 μm . Chromatin condensation in cells is indicated by red arrows, while plasma membrane blebbing is denoted by black arrows. Red rectangles highlight selected cells for magnified images

suppression, triggering growth arrest, apoptosis, and senescence [48]. Furthermore, p53 is instrumental in sensitizing tumor cells to anticancer agents [49]. Given that the gene encoding the p53 tumor suppressor is mutated or altered in many human cancers, it represents a promising targets for mechanism-driven anticancer drug discovery [50]. In studies related to human cancer therapy, the LNCaP, DU-145, and PC-3 cancer cell lines are commonly employed for investigating oncogenes associated with the p53 gene [51]. Specifically, LNCaP harbours a wild-type p53 [52], DU-145 exhibits an inactivating mutation of the p53 gene [53], and PC-3 is a p53-deficient cell line [54]. In this context, these cell lines were utilized to screen the chemotherapeutic potential of *R. undulatum* L. originated aloe-emodin on p53-dependent prostate cancer. Our data indicate that aloe-emodin derived from *R. undulatum* L. suppresses the proliferation of all

three cell lines. Specifically, the viability of LNCaP cells, which possess wild-type p53, was notably reduced following a 48-h treatment with aloe-emodin, whereas the impact on p53-mutant DU-145 and p53-null PC-3 cells was less pronounced (Fig. 2). This observation underscores that the anti-proliferative effect of aloe-emodin on prostate cancer cell types is dependent of the p53 status.

Additionally, the results of protein expression confirmed the successful knockout of the p53 gene in the PC-3 cell line (Fig. 3). PC-3 cells in aloe-emodin treatment groups exhibited no detectable elevation in cleaved-PARP (c-PARP) and displayed no alteration in full-length PARP. Conversely, p53-mutant DU-145 cells demonstrated a comparable caspase-mediated expression of c-PARP and depletion in full-length PARP similar to p53-wild type LNCaP cells (Fig. 3). This observation suggests a direct regulation of PPAR cleavage by the p53 pathway.

The presence of cleaved c-PARP is considered a hallmark of apoptosis induced by chemotherapy [55, 56]. Previous reports have indicated that p53-mutant cell lines show reduced responsiveness to aloe-emodin compared to p53 wild-type cells [57].

Both LNCaP and PC-3 cell lines exhibit high levels of essential AKT survival activity due to PTEN deficiency, while the DU-145 cell line express a lower level of this survival pathway owing to its wild-type PTEN phenotype [58]. PTEN, acting as a tumor suppressor gene, regulates cell cycle arrest, thereby preventing cells from growing, invading, and metastasizing [58, 59]. Consequently, the low sensitivity of DU-145 cells to aloe-emodin-induced apoptosis is likely attributable to its basal survival signaling. These results substantiate the idea that aloe-emodin predominantly triggers apoptosis in LNCaP cells through the p53 pathway. Although previous research has highlighted the induction of apoptosis by chemotherapies through interaction with p53 in LNCaP cells [60], our study stands out by examining the impact of aloe-emodin on p53-dependent apoptosis across three distinct human metastatic prostate cancer cell lines, each with varying p53 statuses. This indicates that the effectiveness of aloe-emodin as an anticancer agent in prostate cancer could hinge on the activity of p53.

Moreover, our investigation revealed that aloe-emodin activated a kinase responsible for phosphorylating p53 at serine 15 (Fig. 4A). Upon activation, the p53 protein initiates caspase-8 cleavage and suppressed Bcl-2. In addition, the Bcl-2-associated X protein (Bax) of Bcl-2 family, which involve in p53-mediated apoptosis, was unaffected by treatment of aloe-emodin. This suggests another cell death mechanism that vary from apoptosis [61]. The apoptosis process induced by p53 was demonstrated to be mediated through both the mitochondrial and the death receptor pathway, involving caspase activation and PARP cleavage [62, 63]. Typically, apoptosis is launched by mitochondrial outer membrane permeabilization (MOMP), defining the mitochondrial or intrinsic pathway, ultimately resulting in caspase initiation and cleavage of protein substrates [64]. MOMP and the following release of pro-apoptotic factors (caspase-8 and 9) can activate effector caspases, which are proteases responsible for breaking down cellular components during apoptosis. By inhibiting MOMP, Bcl-2 contributes to maintaining the integrity of nuclear DNA, protecting it from degradation by these caspases. Additionally, p53 is known to suppress cell survival genes of the Bcl-2 family [65]. The Bax/Bcl-2 ratio represents a switch between cell death and survival with apoptotic stimulations [66]. The higher this ratio is, the more likely cells are failed to survive from environment stimuli. Our findings align with a prior study that similarly demonstrated a programmed

cell death mechanism through p53-mediated caspase activation, PARP cleavage, and reduction of Bcl-2 [67]. However, the cell cycle and proliferation mediator p21, which usually activated by the active p53 [68], express usual protein level with treatment of aloe-emodin. This result suggests a programmed cell death mechanism which is distinct from cell cycle arrest procedure and in a p21-independent pathway.

Figure 4 also illustrates the induction of caspase-8 and caspase-9 cleavage in cells treated with aloe-emodin. Caspase-8 is recognized as a multifaceted protein involved in apoptotic extrinsic pathways triggered by extracellular signals (e.g., Fas ligand), inflammatory responses (via inflammasomes), and it plays a role in the crosstalk between apoptosis and pyroptosis through mitochondrial-activated inflammatory pathways [69–71]. Recent research highlights the central role of caspase-8 as a pivotal switch among programmed cell deaths, including apoptosis, necrosis, and pyroptosis [72]. Enzymatically inactive caspase-8 (C362S) is linked to embryonic lethality due to the induction of necroptosis and pyroptosis. Pyroptosis is triggered by inflammasome in murine intestinal epithelial cells. This programmed cell death involves the direct cleavage of GSDMD by caspase-8, activating macrophages for phagocytosis through innate immune cytokines [73]. Additionally, caspase-9, which was well described as an intrinsic apoptotic initiator, was activated and subsequently induced caspase-3 and GSDME cleavage to mediate the pyroptosis process in response to stimuli [74, 75]. The activation of caspase-9/caspase-3/gasdermin E axis of pyroptosis also be found in Hela cell exposing to aloe-emodin [12]. Therefore, the elevation of both cleaved caspase-8 and caspase-9 in this study give a strong evidence for the switch between apoptosis and pyroptosis cell death mechanism, Pyroptosis leads to DNA fragmentation and cell death, as evidenced by Annexin V staining as well [76]. Besides, aloe-emodin was proved to activate the NF- κ B inflammatory pathway in liver of zebrafish [77]. There might be a correlation between the inflammation and caspase pathways which can lead to pyroptosis. Hence, the activation of cleaved caspase-8 and caspase-9 potentially induces pyroptosis, providing a mechanism to eliminate damaged or cancer cells through the innate immune system.

Annexin V marking of LNCaP cells indicated a dose-dependent elevation in the number of apoptotic cells following an exposure to aloe-emodin for 24 h (Fig. 5). Aloe-emodin has been proved to induce programmed cell death in the wild-type prostate cancer cell line LNCaP within 24 h. Given that aloe-emodin induces DNA damage response in LNCaP cells, Hoechst staining was conducted to assess DNA and nuclear membrane fragmentation (Fig. 6). Aloe-emodin mediated DNA

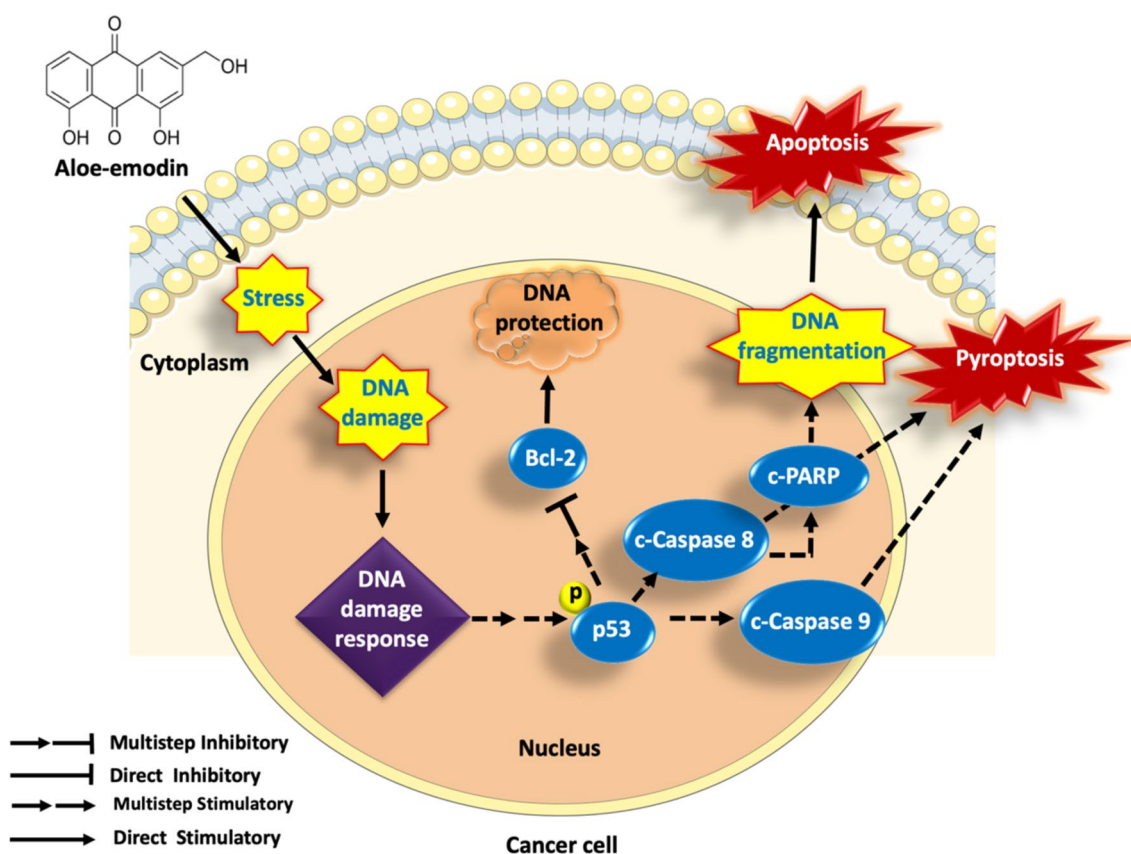


Fig. 7 The anticancer mechanism of aloe-emodin in prostate cancer cell LNCaP

fragmentation and disruption of the nuclear envelope. The activation of DDR involves the formation of DNA fragmentation containing activated H2A.X at uncapped telomeres along with DNA strand breaks. The phosphorylation of histone variant H2A.X (serine 139), referred to as gamma-H2A.X, is essential in checkpoint-mediated cell cycle arrest and DNA repair due to double-stranded breaks processes [78].

Stressed cells also exhibit disrupted nuclear envelope integrity resulting from the reduced expression of lamin B1, a major trigger of cellular senescence [79]. We previously examined nuclear membrane fragmentation through certain markers such as the expression of lamin-B1 and p-H2A.X. Additionally, we conducted assays to evaluate the effects of aloe-emodin treatment at various doses (0, 20, 50, 100 μ M) on LNCaP cells. We assessed senescence-associated β -galactosidase activity and cell morphology after 48 h of treatment. At 20 μ M dose of aloe-emodin, the flattened cell morphology and significant fluorescent accumulation were also observed after 24 h exposure, indicating cellular senescence.

Furthermore, we performed cell proliferation assays with aloe-emodin at different doses (0, 20, 50, 100 μ M) over 4 and 8 days. The results showed that 20 μ M dose caused a permanent cell cycle arrest, while the 50 and 100 μ M doses led to cell death [45]. These findings suggest that low and high doses of aloe-emodin produce distinct effects on the prostate cancer-derived LNCaP cell line.

Regarding to DNA damage, activation of p53 enhances its DNA-binding ability, leading to cell cycle arrest, apoptosis, or DNA repair [80]. Aloe-emodin might also mediated DNA damage response by inducing serine 139 phosphorylation in histone H2A.X in LNCaP cells [81]. Our findings address distinct p53-dependent programmed cell death pathways compared to those reported in previous studies on DNA damage response to aloe-emodin in human colorectal adenocarcinoma (WiDr), human gastric cancer (AGS and NCI-N87) [82], human lung squamous cell carcinoma (CH27) as well as human lung non-small cell carcinoma 7(H460) [83].

In conclusion, aloe-emodin isolated from *R. undulatum* L. effectively inhibited the proliferation of human

prostate cancer LNCaP cells by triggering p53-mediated programmed cell death and impeding DNA protection. Aloe-emodin initiated apoptosis through the activation of p53 signaling, featuring key indicators, such as caspase-8 activation, PARP cleavage, Bcl-2 suppression, and DNA fragmentation (Fig. 7). The anticancer effect of aloe-emodin-induced p53-mediated cellular senescence at lower doses could potentially serve as an adjunctive approach in the early stages of prostate cancer treatment, while aloe-emodin-induced p53-mediated apoptosis might be considered for the later stages of the treatment progression. Additionally, the cleavage of caspase-8, marking not only the apoptotic process but also initiating pyroptosis, represents a novel direction in cancer therapy. In our study, we found DNA fragmentation by Annexin V staining and an induction of cleaved caspase-8 and 9 proteins in cells treated with aloe-emodin. These factors can also be found in pyroptosis process as well. This study contributes to understanding the pharmacological potential of aloe-emodin, providing a foundation for further exploration of its anticancer pathways in human cancers and in the immune system.

Abbreviations

<i>R. undulatum</i>	<i>Rheum undulatum</i> L.
mTOR	Mammalian target of rapamycin
p53	Tumor protein P53
TNF	Tumor necrosis factor
VEGF	Vascular endothelial growth factor
MAPKs	Mitogen-activated protein kinases
PI3K/AKT pathway	Phosphatidylinositol 3'-kinase (PI3K)-Akt pathway
Akt	Protein kinase B
ATM	ATM Serine/Threonine kinase
ATR	Serine/threonine-protein kinase ATR
BRCA1	Breast cancer type 1 susceptibility protein
MGMT	Methylated-DNA-protein-cysteine methyltransferase
ERK	Extracellular signal-regulated kinases
ALP	Alkaline phosphatase
PKC pathway	Protein kinase C pathway
Fas/APO1	Fas receptor/apoptosis antigen 1
Bax	BCL2 associated X (apoptosis regulator)
p21	Cyclin-dependent kinase inhibitor p21
CAPN2	Calpain 2
UBE3A	Ubiquitin protein ligase E3A
ERα	Estrogen receptor alpha
NAT enzyme	N-acetyltransferase
MMP	Matrix metalloproteinase
NF-κB	Nuclear factor kappa B family transcription factor
MAPK/JNK	Mitogen-activated protein kinase/ c-Jun N-terminal kinases
FAK	Focal adhesion kinase
p38 MAPKs	Proline-directed serine/threonine kinases of MAPKs
ROS-JNK	Reactive oxygen species activated JNK
SMAC/DIABLO	Second mitochondria-derived activator of caspase/ direct inhibitor of apoptosis-binding protein with low pI
TAK1	Mitogen-activated protein kinase kinase kinase 7
p73	Tumor protein P73
MeOH	Methanol
CHCl ₃	Chloroform
EtOAc	Ethyl acetate
RPMI 1640	Roswell Park Memorial Institute 1640
MEM	Essential medium
FBS	Fetal bovine serum

P/S	Penicillin/streptomycin
OD	Optical density
IC ₅₀	Half maximal inhibitory concentration
DPBS	Dulbecco's phosphate-buffered saline
SDS-PAGE	Sodium dodecyl sulfate-polyacrylamide gel electrophoresis
SEM	Standard error of the mean
GSK3β	Glycogen synthase kinase-3 beta
HSP90αA1	Heat shock protein 90 alpha family class A member 1
NOS2	Nitric oxide synthase 2
PTGS2	Prostaglandin-endoperoxide synthase 2
ESR1-2	Estrogen receptor 1-2; CASP (3-9): Caspase (3-9)
AR	Androgen receptor
CDK 1-2	Cyclin-dependent kinases 1-2
PIM1	Pim-1 proto-oncogene, serine/threonine kinase
PPARγ	Peroxisome proliferator-activated receptor gamma
PRKACα	Protein kinase CAMP-activated catalytic subunit alpha
CHEK1	Checkpoint kinase 1
IL1β	Interleukin 1 beta
PTGS2	Prostaglandin-endoperoxide synthase 2
NOS3	Nitric oxide synthase 3
EIF2S1	Eukaryotic translation initiation factor 2 subunit 1
CCNA2	Cyclin-A2
NCOA2	Nuclear receptor coactivator 2
HR-ESI-MS	High-resolution electrospray ionization mass spectrometry
NMR	Nuclear magnetic resonance
c-PARP	Cleaved Poly (ADP-ribose) polymerase
Bcl-2	B-cell lymphoma 2
DDR	DNA damage response
PTEN	Phosphatase and tensin homolog
MOMP	Mitochondrial outer membrane permeabilization
C362S	Enzymatically inactive caspase-8
H2A.X	H2A histone family member X

Acknowledgements

Not applicable.

Author contributions

Conceptualization, K. S. K. and J. C.; methodology, N. K. S. T., N. Q. N., and S. L.; formal analysis, S. H. K., D. J., E. S., and J. J. P.; writing original draft preparation, N. K. S. T. and N. Q. N.; writing review and editing, Ki Sung Kang and J. C.; project administration, K. S. K.; funding acquisition, K. S. K. All authors have read and agreed to the published version of the manuscript.

Funding

This research was supported by the Bio & Medical Technology Development Program of the National Research Foundation (NRF) funded by the Ministry of Science & ICT (2020M3A9E4103843). This work was also supported by the National Foundation of Korea (NRF), funded by the Ministry of Science, ICT & Future Planning (2017M3A9133061313).

Availability of data and materials

All data generated or analysed during this study are included in this published article [and its supplementary information files].

Declarations

Competing interests

The authors declare no competing interests. Ki Sung Kang is an Associate Editor of Applied Biological Chemistry. Associate Editor status has no bearing on editorial consideration.

Author details

¹NTT Hi-Tech Institute, Nguyen Tat Thanh University, District 09, Ho Chi Minh City 70000, Vietnam. ²College of Korean Medicine, Gachon University, Seongnam 13120, Republic of Korea. ³Department of Life Science, College of Bio-Nano Technology, Gachon University, Seongnam 13120, Republic of Korea. ⁴College of Pharmacy, Yonsei Institute of Pharmaceutical Sciences, Yonsei University, Incheon 21983, Republic of Korea. ⁵National Center of Excellence

in Software, Faculty of SW Convergence, College of Convergence Engineering, Sangmyung University, Seoul, Republic of Korea. ⁶Department of Dental Regenerative Biotechnology, School of Dentistry, Seoul National University, Seoul, Republic of Korea. ⁷Dental Research Institute, Seoul National University, Seoul, Republic of Korea.

Received: 11 July 2024 Accepted: 13 November 2024

Published online: 11 December 2024

References

- Jarvis C (2007) Order out of chaos. Linnaean plant names and their types, 1st edn. The Linnean Society of London in association with the Natural History Museum, London
- Choi SZ, Lee SO, Jang KU, Chung SH, Park SH, Kang HC, Yang EY, Cho HJ, Lee KR (2005) Antidiabetic stilbene and anthraquinone derivatives from *rheum undulatum*. *Arch Pharm Res* 28:1027–1030
- Matsuda H, Tewtrakul S, Morikawa T, Yoshikawa M (2004) Anti-allergic activity of stilbenes from korean rhubarb (*rheum undulatum* L.): Structure requirements for inhibition of antigen-induced degranulation and their effects on the release of *tnf-α* and *il-4* in *rbl-2h3* cells. *Bioorg Med Chem* 12:4871–4876
- Wang L, Pan X, Sweet DH (2013) The anthraquinone drug rhein potentially interferes with organic anion transporter-mediated renal elimination. *Biochem Pharmacol* 86:991–996
- Ha MT, Park DH, Shrestha S, Kim M, Kim JA, Woo MH, Choi JS, Min BS (2018) Ptp1b inhibitory activity and molecular docking analysis of stilbene derivatives from the rhizomes of *rheum undulatum* L. *Fitoterapia* 131:119–126
- Siddamurthi S, Gutti G, Jana S, Kumar A, Singh SK (2020) Anthraquinone: a promising scaffold for the discovery and development of therapeutic agents in cancer therapy. *Future Med Chem* 12:1037–1069
- Kim J-E, Kim H-J, Pandit S, Chang K-W, Jeon J-G (2011) Inhibitory effect of a bioactivity-guided fraction from *rheum undulatum* on the acid production of *streptococcus mutans* biofilms at sub-mic levels. *Fitoterapia* 82:352–356
- Hu J, Lloyd M, Hobbs C, Cox P, Burke K, Pearce G, Streicker MA, Gao Q, Frankos V (2021) Absence of genotoxicity of purified aloe vera whole leaf dry juice as assessed by an in vitro mouse lymphoma tk assay and an in vivo comet assay in male f344 rats. *Toxicol Rep* 8:511–519
- Akbar S, Akbar S (2020) *Rheum emodi* wall. Ex meism (polygonaceae) (syn.: *Rheum australe* d. Don). Handbook of 200 Medicinal plants: a comprehensive review of their traditional medical uses and scientific justifications. Springer, Cham, pp 1521–1526
- Dong X, Zeng Y, Liu Y, You L, Yin X, Fu J, Ni J (2020) Aloe-emodin: a review of its pharmacology, toxicity, and pharmacokinetics. *Phytother Res* 34:270–281
- Sanders B, Ray AM, Goldberg S, Clark T, McDaniel HR, Atlas SE et al (2018) Anti-cancer effects of aloe-emodin: a systematic review. *J Clin Transl Res* 3:283
- Li T, Shi L, Liu W, Hu X, Hui Y, Di M et al (2022) Aloe-emodin induces mitochondrial dysfunction and pyroptosis by activation of the caspase-9/3/gasdermin e axis in hela cells. *Front Pharmacol* 13:854526
- Liu K, Park C, Li S, Lee KW, Liu H, He L et al (2012) Aloe-emodin suppresses prostate cancer by targeting the mtor complex 2. *Carcinogenesis* 33:1406–1411
- Guo J, Xiao B, Liu Q, Gong Z, Le Y (2008) Suppression of c-myc expression associates with anti-proliferation of aloe-emodin on gastric cancer cells. *Cancer Invest* 26:369–374
- Chen H, Hsieh W, Chang W, Chung J (2004) Aloe-emodin induced in vitro g2/m arrest of cell cycle in human promyelocytic leukemia hl-60 cells. *Food Chem Toxicol* 42:1251–1257
- Suboj P, Babykutty S, Srinivas P, Gopala S (2012) Aloe emodin induces g2/m cell cycle arrest and apoptosis via activation of caspase-6 in human colon cancer cells. *Pharmacology* 89:91–98
- Luo J, Yuan Y, Chang P, Li D, Liu Z, Qu Y (2014) Combination of aloe-emodin with radiation enhances radiation effects and improves differentiation in human cervical cancer cells. *Mol Med Rep* 10:731–736
- Suhartati T, Sadiha K, Yandri Y, Hadi S (2023) Anticancer activity study of modified artocarpin compound from puda plant (*artocarpus kemando* miq.). *Emerg Sci J* 7:733–743
- Suhartati T, Prihatin AS, Kurniati AN, Ropongi H, Yandri Y, Hadi S (2024) Cycloartobioxanthone, a flavonoid with antidiabetic, antibacterial and anticancer activities from *artocarpus kemando* miq. *Emerg Sci J* 8:43–60
- Roshani S, Coccia M, Mosleh M (2022) Sensor technology for opening new pathways in diagnosis and therapeutics of breast, lung, colorectal and prostate cancer. *medRxiv:2022.2002.2018.22271186*
- Hodson R (2015) Small organ, big reach. *Nature* 528:S118
- Law A, McLaren D (2010) Non-surgical treatment for early prostate cancer. *J R Coll Phys Edinb* 40:340–342
- Bubendorf L, Schöpfer A, Wagner U, Sauter G, Moch H, Willi N, Gasser TC, Mihatsch MJ (2000) Metastatic patterns of prostate cancer: an autopsy study of 1,589 patients. *Hum Pathol* 31:578–583
- Lawrentschuk N, Trotter G, Kuk C, Zlotta A (2010) Role of surgery in high-risk localized prostate cancer. *Curr Oncol* 17:25–32
- O'Hanlon Brown C, Waxman J (2012) Current management of prostate cancer: dilemmas and trials. *Br J Radiol* 85:S28–S40
- Hanahan D, Weinberg RA (2011) Hallmarks of cancer: the next generation. *Cell* 144:646–674
- Peng F, Liao M, Qin R, Zhu S, Peng C, Fu L, Chen Y, Han B (2022) Regulated cell death (rkd) in cancer: key pathways and targeted therapies. *Signal Transduct Target Ther* 7:286
- Sever AI, Alderson TR, Rennella E, Aramini JM, Liu ZH, Harkness RW, Kay LE (2023) Activation of caspase-9 on the apoptosome as studied by methyl-trosy nmr. *Proc Natl Acad Sci U S A* 120:e2310944120
- Liu J, Uematsu H, Tsuchida N, Ikeda M-A (2011) Essential role of caspase-8 in p53/p73-dependent apoptosis induced by etoposide in head and neck carcinoma cells. *Mol Cancer* 10:1–13
- Sarhan J, Liu BC, Muendlein HI, Li P, Nilson R, Tang AY et al (2018) Caspase-8 induces cleavage of gasdermin d to elicit pyroptosis during yersinia infection. *Proc Natl Acad Sci U S A* 115:E10888–E10897
- Wei X, Xie F, Zhou X, Wu Y, Yan H, Liu T et al (2022) Role of pyroptosis in inflammation and cancer. *Cell Mol Immunol* 19:971–992
- Perkins CL, Fang G, Kim CN, Bhalla KN (2000) The role of apaf-1, caspase-9, and bid proteins in etoposide- or paclitaxel-induced mitochondrial events during apoptosis. *Cancer Res* 60:1645–1653
- Müller I, Strozzyk E, Schindler S, Beissert S, Oo HZ, Sauter T et al (2020) Cancer cells employ nuclear caspase-8 to overcome the p53-dependent g2/m checkpoint through cleavage of usp28. *Mol Cell* 77(970–984):e977
- Zhou B, Zhang J-y, Liu X-s, Chen H-z, Ai Y-l, Cheng K et al (2018) Tom20 senses iron-activated ros signaling to promote melanoma cell pyroptosis. *Cell Res* 28:1171–1185
- Hou J, Zhao R, Xia W, Chang C-W, You Y, Hsu J-M et al (2020) Pd-I1-mediated gasdermin c expression switches apoptosis to pyroptosis in cancer cells and facilitates tumour necrosis. *Nat Cell Biol* 22:1264–1275
- Wan J, Zhang J, Zhang J (2018) Expression of p53 and its mechanism in prostate cancer. *Oncol Lett* 16:378–382
- Harris SL, Levine AJ (2005) The p53 pathway: positive and negative feedback loops. *Oncogene* 24:2899–2908
- Rufini A, Tucci P, Celardo I, Melino G (2013) Senescence and aging: the critical roles of p53. *Oncogene* 32:5129–5143
- Masuda Y, Futamura M, Kamino H, Nakamura Y, Kitamura N, Ohnishi S et al (2006) The potential role of dfna5, a hearing impairment gene, in p53-mediated cellular response to DNA damage. *J Hum Genet* 51:652–664
- Özener N, Saeed M, Demirezer LÖ, Efferth T (2018) Aloe-emodin as drug candidate for cancer therapy. *Oncotarget* 9:17770
- Huh E, Choi JG, Choi Y, Ju IG, Kim B, Shin YJ, An JM, Park MG, Yim SV, Chung SJ, Seo SU, Kim D, Kim CH, Kim DH, Oh MS (2023) P. mirabilis-derived pore-forming haemolysin, HpmA drives intestinal alpha-synuclein aggregation in a mouse model of neurodegeneration. *EBioMedicine* 98:104887
- Jo HG, Seo J, Lee D (2022) Clinical evidence construction of East Asian herbal medicine for inflammatory pain in rheumatoid arthritis based on integrative data mining approach. *Pharmacol Res* 185:106460
- Aoki-Kinoshita KF, Kanehisa M (2007) Gene annotation and pathway mapping in kegg. *Comparative genomics*: 71–91

44. Lee D-H, Kwak HJ, Shin Y, Kim SJ, Lee GH, Park I-H, Kim SH, Kang KS (2023) Elucidation of phytochemicals affecting platelet responsiveness in dangguisu-san: active ingredient prediction and experimental research using network pharmacology. *Plants* 12:1120
45. He MT, Nguyen QN, Cho EJ, Kim SH, Park S, Park JY, Lee S, Kang KS (2024) Aloe-emodin isolated from rheum undulatum L. Regulates cell cycle distribution and cellular senescence in human prostate cancer Incap cells. *J Diet Suppl* 21:389–407
46. Rawla P (2019) Epidemiology of prostate cancer. *World J Oncol* 10:63
47. Thornton AA, Perez MA (2006) Posttraumatic growth in prostate cancer survivors and their partners. *Psychooncology* 15:285–296
48. Hientz K, Mohr A, Bhakta-Guha D, Efferth T (2017) The role of p53 in cancer drug resistance and targeted chemotherapy. *Oncotarget* 8:8921
49. Sax J, El-Deiry W (2003) P53 downstream targets and chemosensitivity. *Cell Death Differ* 10(4):413–7
50. Koster R, Timmer-Bosscha H, Bischoff R, Gietema JA, de Jong S (2011) Disruption of the mdm2–p53 interaction strongly potentiates p53-dependent apoptosis in cisplatin-resistant human testicular carcinoma cells via the fas/fasL pathway. *Cell Death Dis* 2:e148–e148
51. Carroll AG, Voeller HJ, Sugars L, Gelmann EP (1993) P53 oncogene mutations in three human prostate cancer cell lines. *Prostate* 23:123–134
52. Narayanan BA, Narayanan NK, Re GG, Nixon DW (2003) Differential expression of genes induced by resveratrol in Incap cells: P53-mediated molecular targets. *Int J Cancer* 104:204–212
53. Lecane PS, Kiviharju TM, Sellers RG, Peehl DM (2003) Leptomycin b stabilizes and activates p53 in primary prostatic epithelial cells and induces apoptosis in the Incap cell line. *Prostate* 54:258–267
54. Isaacs WB, Carter BS, Ewing CM (1991) Wild-type p53 suppresses growth of human prostate cancer cells containing mutant p53 alleles. *Cancer Res* 51:4716–4720
55. Kaufmann SH (1989) Induction of endonucleolytic DNA cleavage in human acute myelogenous leukemia cells by etoposide, camptothecin, and other cytotoxic anticancer drugs: a cautionary note. *Cancer Res* 49:5870–5878
56. Kaufmann SH, Desnoyers S, Ottaviano Y, Davidson NE, Poirier GG (1993) Specific proteolytic cleavage of poly (adp-ribose) polymerase: an early marker of chemotherapy-induced apoptosis. *Cancer Res* 53:3976–3985
57. Pecere T, Sarinella F, Salata C, Gatto B, Bet A, Dalla Vecchia F et al (2003) Involvement of p53 in specific anti-neuroectodermal tumor activity of aloe-emodin. *Int J Cancer* 106:836–847
58. Li J, Yen C, Liaw D, Podsypanina K, Bose S, Wang SI et al (1997) Pten, a putative protein tyrosine phosphatase gene mutated in human brain, breast, and prostate cancer. *Science* 275:1943–1947
59. Song MS, Salmena L, Pandolfi PP (2012) The functions and regulation of the pten tumour suppressor. *Nat Rev Mol Cell Biol* 13:283–296
60. Zhang Z-W, Yang Z-M, Zheng Y-C, Chen Z-D (2010) Transgelin induces apoptosis of human prostate Incap cells through its interaction with p53. *Asian J Androl* 12:186
61. Aghaei M, KhanAhmad H, Aghaei S, Nilforoushzadeh MA, Mohaghegh M-A, Hejazi SH (2020) The role of bax in the apoptosis of leishmania-infected macrophages. *Microb Pathog* 139:103892
62. Green DR, Kroemer G (2009) Cytoplasmic functions of the tumour suppressor p53. *Nature* 458:1127–1130
63. Levine AJ (1997) P53, the cellular gatekeeper for growth and division. *Cell* 88:323–331
64. Kalkavan H, Green DR (2018) Momp, cell suicide as a bcl-2 family business. *Cell Death Differ* 25:46–55
65. Riley T, Sontag E, Chen P, Levine A (2008) Transcriptional control of human p53-regulated genes. *Nat Rev Mol Cell Biol* 9:402–412
66. Gao C, Wang A-Y (2009) Significance of increased apoptosis and bax expression in human small intestinal adenocarcinoma. *J Histochem Cytochem* 57:1139–1148
67. Hastak K, Gupta S, Ahmad N, Agarwal MK, Agarwal ML, Mukhtar H (2003) Role of p53 and nf-kb in epigallocatechin-3-gallate-induced apoptosis of Incap cells. *Oncogene* 22:4851–4859
68. Abbas T, Dutta A (2009) P21 in cancer: intricate networks and multiple activities. *Nat Rev Cancer* 9:400–414
69. Boatright KM, Salvesen GS (2003) Mechanisms of caspase activation. *Curr Opin Chem Biol* 15:725–731
70. Galluzzi L, Vitale I, Aaronson SA, Abrams JM, Adam D, Agostinis P et al (2018) Molecular mechanisms of cell death: recommendations of the nomenclature committee on cell death 2018. *Cell Death Differ* 25:486–541
71. Vringer E, Tait SW (2023) Mitochondria and cell death-associated inflammation. *Cell Death Differ* 30:304–312
72. Fritsch M, Günther SD, Schwarzer R, Albert M-C, Schorn F, Werthenbach JP et al (2019) Caspase-8 is the molecular switch for apoptosis, necroptosis and pyroptosis. *Nature* 575:683–687
73. Orning P, Weng D, Starheim K, Ratner D, Best Z, Lee B et al (2018) Pathogen blockade of tak1 triggers caspase-8–dependent cleavage of gasdermin d and cell death. *Science* 362:1064–1069
74. Aral K, Aral CA, Kapila Y (2019) The role of caspase-8, caspase-9, and apoptosis inducing factor in periodontal disease. *J Periodontol* 90:288–294
75. Lu H, Zhang S, Wu J, Chen M, Cai M-C, Fu Y et al (2018) Molecular targeted therapies elicit concurrent apoptotic and gsdme-dependent pyroptotic tumor cell death. *Clin Cancer Res* 24:6066–6077
76. Siegel RM (2006) Caspases at the crossroads of immune-cell life and death. *Nat Rev Immunol* 6:308–317
77. Quan Y, Gong L, He J, Zhou Y, Liu M, Cao Z, Li Y, Peng C (2019) Aloe emodin induces hepatotoxicity by activating nf-kb inflammatory pathway and p53 apoptosis pathway in zebrafish. *Toxicol Lett* 306:66–79
78. Yuan J, Adamski R, Chen J (2010) Focus on histone variant h2ax: to be or not to be. *FEBS Lett* 584:3717–3724
79. Wang C, Jurk D, Maddick M, Nelson G, Martin-Ruiz C, Von Zglinicki T (2009) DNA damage response and cellular senescence in tissues of aging mice. *Aging Cell* 8:311–323
80. Lakin ND, Jackson SP (1999) Regulation of p53 in response to DNA damage. *Oncogene* 18:7644–7655
81. Devlin H-L, Mack PC, Burich RA, Gumerlock PH, Kung H-J, Mudryj M, deVere White RW (2008) Impairment of the DNA repair and growth arrest pathways by p53r2 silencing enhances DNA damage-induced apoptosis in a p53-dependent manner in prostate cancer cells. *Mol Cancer Res* 6:808–818
82. Chen S-H, Lin K-Y, Chang C-C, Fang C-L, Lin C-P (2007) Aloe-emodin-induced apoptosis in human gastric carcinoma cells. *Food Chem Toxicol* 45:2296–2303
83. Lee HZ (2001) Protein kinase c involvement in aloe-emodin-and emodin-induced apoptosis in lung carcinoma cell. *Br J Pharmacol* 134:1093–1103
84. Lee H-Z, Hsu S-L, Liu M-C, Wu C-H (2001) Effects and mechanisms of aloe-emodin on cell death in human lung squamous cell carcinoma. *Eur J Pharmacol* 431:287–295
85. Yeh FT, Wu CH, Lee HZ (2003) Signaling pathway for aloe-emodin-induced apoptosis in human h460 lung nonsmall carcinoma cell. *Int J Cancer* 106:26–33
86. Lai M-Y, Hour M-J, Leung HW-C, Yang W-H, Lee H-Z (2007) Chaperones are the target in aloe-emodin-induced human lung nonsmall carcinoma h460 cell apoptosis. *Eur J Pharmacol* 573:1–10
87. Wu Y-Y, Zhang J-H, Gao J-H, Li Y-S (2017) Aloe-emodin (ae) nanoparticles suppresses proliferation and induces apoptosis in human lung squamous carcinoma via ros generation in vitro and in vivo. *Biochem Biophys Res Commun* 490:601–607
88. Wei W-T, Lin S-Z, Liu D-L, Wang Z-H (2013) The distinct mechanisms of the antitumor activity of emodin in different types of cancer. *Oncol Rep* 30:2555–2562
89. Chang X, Zhao J, Tian F, Jiang Y, Lu J, Ma J et al (2016) Aloe-emodin suppresses esophageal cancer cell te1 proliferation by inhibiting akt and erk phosphorylation. *Oncol Lett* 12:2232–2238
90. Şeker Karatoprak G, Küpeli Akkol E, Yücel Ç, Bahadır Acikara Ö, Sobarzo-Sánchez E (2022) Advances in understanding the role of aloe emodin and targeted drug delivery systems in cancer. *Oxid Med Cell Longev* 2022:7928200
91. Guo J, Xiao B, Zhang S, Liu D, Liao Y, Sun Q (2007) Growth inhibitory effects of gastric cancer cells with an increase in s phase and alkaline phosphatase activity repression by aloe-emodin. *Cancer Biol Ther* 6:85–88
92. Chihara T, Shimpo K, Beppu H, Yamamoto N, Kaneko T, Wakamatsu K, Sonoda S (2015) Effects of aloe-emodin and emodin on proliferation

- of the mkn45 human gastric cancer cell line. *Asian Pac J Cancer Prev* 16:3887–3891
93. Kuo P-L, Lin T-C, Lin C-C (2002) The antiproliferative activity of aloe-emodin is through p53-dependent and p21-dependent apoptotic pathway in human hepatoma cell lines. *Life Sci* 71:1879–1892
94. Lu GD, Shen H-M, Chung MC, Ong CN (2007) Critical role of oxidative stress and sustained jnk activation in aloe-emodin-mediated apoptotic cell death in human hepatoma cells. *Carcinogenesis* 28:1937–1945
95. Lu GD, Shen HM, Ong CN, Chung MC (2007) Anticancer effects of aloe-emodin on hepg2 cells: cellular and proteomic studies. *Proteomics Clin Appl* 1:410–419
96. Lin ML, Lu YC, Su HL, Lin HT, Lee CC, Kang SE, Lai TC, Chung JG, Chen SS (2011) Destabilization of carp mrnas by aloe-emodin contributes to caspase-8-mediated p53-independent apoptosis of human carcinoma cells. *J Cell Biochem* 112:1176–1191
97. Jeon W, Jeon YK, Nam MJ (2012) Apoptosis by aloe-emodin is mediated through down-regulation of calpain-2 and ubiquitin-protein ligase e3a in human hepatoma huh-7 cells. *Cell Biol Int* 36:163–167
98. Chen Q, Tian S, Zhu J, Li K-T, Yu T-H, Yu L-H, Bai D-Q (2016) Exploring a novel target treatment on breast cancer: aloe-emodin mediated photodynamic therapy induced cell apoptosis and inhibited cell metastasis. *Anticancer Agents Med Chem* 16:763–770
99. Huang P-H, Huang C-Y, Chen M-C, Lee Y-T, Yue C-H, Wang H-Y, Lin H (2013) Emodin and aloe-emodin suppress breast cancer cell proliferation through era inhibition. *Evid Based Complement Alternat Med* 2013:2013:376123
100. Tseng H-S, Wang Y-F, Tzeng Y-M, Chen D-R, Liao Y-F, Chiu H-Y, Hsieh W-T (2017) Aloe-emodin enhances tamoxifen cytotoxicity by suppressing ras/erk and pi3k/mtor in breast cancer cells. *Am J Chin Med* 45:337–350
101. Ma J-W, Hung C-M, Lin Y-C, Ho C-T, Kao J-Y, Way T-D (2016) Aloe-emodin inhibits her-2 expression through the downregulation of y-box binding protein-1 in her-2-overexpressing human breast cancer cells. *Oncotarget* 7:58915
102. Lin S-Y, Yang J-H, Hsia T-C, Lee J-H, Chiu T-H, Wei Y-H, Chung J-G (2005) Effect of inhibition of aloe-emodin on n-acetyltransferase activity and gene expression in human malignant melanoma cells (a375. S2). *Melanoma Res* 15:489–494
103. Radovic J, Maksimovic-Ivanic D, Timotijevic G, Popadic S, Ramic Z, Trajkovic V, Miljkovic D, Stosic-Grujicic S, Mijatovic S (2012) Cell-type dependent response of melanoma cells to aloe emodin. *Food Chem Toxicol* 50:3181–3189
104. Tabolacci C, Lentini A, Mattioli P, Provenzano B, Oliverio S, Carlomosti F, Beninati S (2010) Antitumor properties of aloe-emodin and induction of transglutaminase 2 activity in b16-f10 melanoma cells. *Life Sci* 87:316–324
105. Tabolacci C, Cordella M, Turcano L, Rossi S, Lentini A, Mariotti S et al (2015) Aloe-emodin exerts a potent anticancer and immunomodulatory activity on braf-mutated human melanoma cells. *Eur J Pharmacol* 762:283–292
106. Tabolacci C, Oliverio S, Lentini A, Rossi S, Galbiati A, Montesano C et al (2011) Aloe-emodin as antiproliferative and differentiating agent on human u937 monoblastic leukemia cells. *Life Sci* 89:812–820
107. Suboj P, Babykutty S, Gopi DRV, Nair RS, Srinivas P, Gopala S (2012) Aloe emodin inhibits colon cancer cell migration/angiogenesis by down-regulating mmp-2/9, rhob and vegf via reduced DNA binding activity of nf-kb. *Eur J Pharm Sci* 45:581–591
108. Shimpo K, Chihara T, Kaneko T, Beppu H, Wakamatsu K, Shinzato M, Yukitake J, Sonoda S (2014) Inhibitory effects of low-dose aloe-emodin on the development of colorectal tumors in min mice. *Asian Pac J Cancer Prev* 15:5587–5592
109. Guo J-m, Xiao B-x, Liu Q, Zhang S, Liu D-h, Gong Z-h (2007) Anticancer effect of aloe-emodin on cervical cancer cells involves g2/m arrest and induction of differentiation. *Acta Pharmacol Sin* 28:1991–1995
110. Acevedo-Duncan M, Russell C, Patel S, Patel R (2004) Aloe-emodin modulates pkc isozymes, inhibits proliferation, and induces apoptosis in u-373mg glioma cells. *Int Immunopharmacol* 4:1775–1784
111. Mijatovic S, Maksimovic-Ivanic D, Radovic J, Miljkovic D, Harhaji L, Vuckovic O, Stosic-Grujicic S, Mostarica Stojkovic M, Trajkovic V (2005) Anti-glioma action of aloe emodin: the role of erk inhibition. *Cell Mol Life Sci* 62:589–598
112. Ismail S, Haris K, Abdul Ghani ARI, Abdullah JM, Johan MF, Mohamed Yusoff AA (2013) Enhanced induction of cell cycle arrest and apoptosis via the mitochondrial membrane potential disruption in human u87 malignant glioma cells by aloe emodin. *J Asian Nat Prod Res* 15:1003–1012
113. He T-P, Yan W-H, Mo L-E, Liang N-C (2008) Inhibitory effect of aloe-emodin on metastasis potential in ho-8910pm cell line. *J Asian Nat Prod Res* 10:383–390
114. Lin J-G, Chen G-W, Li T-M, Chouh S-T, Tan T-W, Chung J-G (2006) Aloe-emodin induces apoptosis in t24 human bladder cancer cells through the p53 dependent apoptotic pathway. *J Urol* 175:343–347
115. 林孟亮, 呂耀呈, 陳世順 (2010) Aloe-emodin induces apoptosis of human nasopharyngeal carcinoma cells via caspase-8-mediated activation of the mitochondrial death pathway
116. Tu P, Huang Q, Ou Y, Du X, Li K, Tao Y, Yin H (2016) Aloe-emodin-mediated photodynamic therapy induces autophagy and apoptosis in human osteosarcoma cell line mg-63 through the ros/jnk signaling pathway. *Oncol Rep* 35:3209–3215

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.